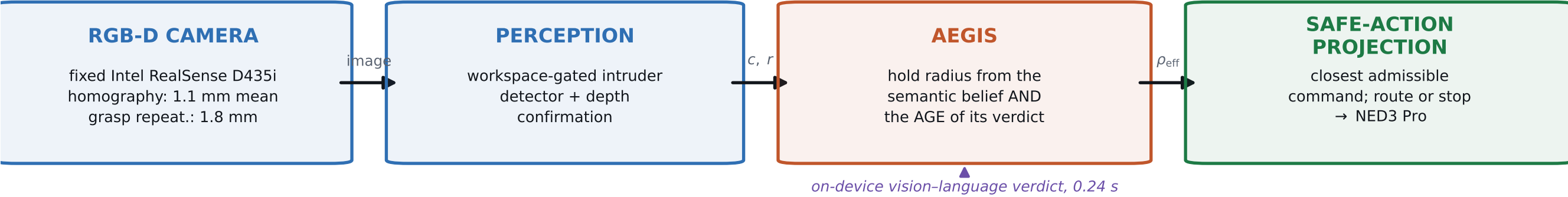
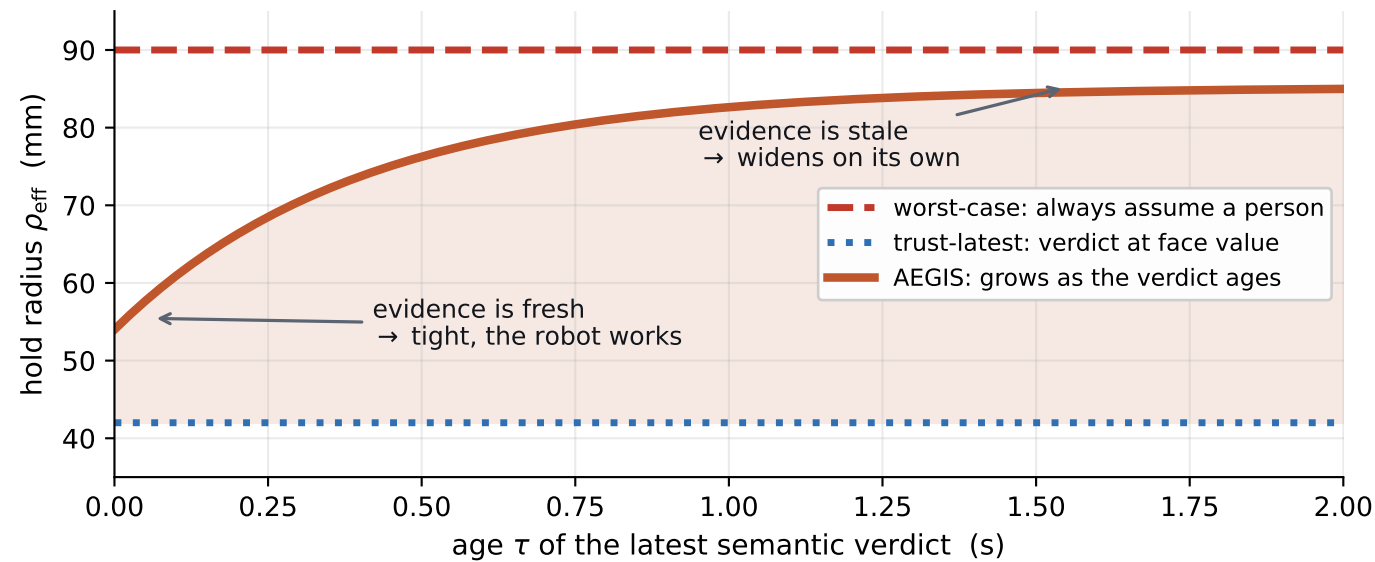


AEGIS — a safety margin that knows how old its evidence is

RGB-D sensing + vision-language monitoring, validated closed-loop on a low-cost manipulator



The mechanism: the margin is a function of verdict age



Measured on the physical robot

worst-case	never delivers with the hazard present	0 ticks of motion near the hazard
trust-latest	delivers	enters the obstacle footprint (−5.6 mm)
AEGIS	delivers	margin never negative (worst +19.7 mm)

one pipeline, one scene, one carry — only the radius rule differs

0.24 s

on-device semantic verdict
100% hazard recall, 0 false stops

1.1 mm

image-to-table homography
residual (2.4 mm worst)

30/30

autonomous carries delivered
on the physical arm

3-29×

unsafe-cost reduction in
simulation, success unchanged